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Quentin Rebergue - Graduate Software Engineer

Profile

Looking for a full-time position in software engineering with a focus on low-level systems / hardware integration.

MSc in Computer Engineering (EPITA - GISTRE) in Systems / Embedded.

Fluent in French and English (TOEIC 915, C1). *Interests: watchmaking, music, hiking/running, cooking.*

Work Experience

- **Head of Teaching Assistants — YAKA / ACU Program, EPITA** 09/2024 – 02/2026

  Led the national team of 67 TAs and supervised 5 intensive coding bootcamps (Java, C++, JS, C/Bash, SQL) and 7 large-scale projects (compiler, HTTP server in C, malloc, POSIX shell) for 700+ 3rd-year students across 5 campuses.

- **Web Developer, Various Clients** 09/2021 – 03/2024

  Designed, deployed, managed information system and apps for 10+ client missions.

- **Software Engineer Intern — Platform team, ARISTA** 02/2026 – 08/2026

  Implemented non-zero reset value management for ASIC Physical Table registers in the chip-agnostic infrastructure. Won 1st place at Arista's internal hackathon by building an AI agent optimization tool (-70% token usage).

School Projects

- **Distributed Task Queue** (Python, Redis, Docker, FastAPI) – Designed and implemented a distributed task queue system with Redis as the broker. Supports retries, dead-letter queues, and priority scheduling. Deployed with Docker Compose.
- **RFID Driver** (C, Linux Kernel, SPI, QEMU) – Developed a Linux kernel character device driver for the MFRC522 RFID reader with file_operations callbacks, SPI regmap communication, and a FIFO command interface. Tested via QEMU emulation.
- **System Analysis 1 - Tracing** (Bash, strace, Linux, System Profiling) – System analysis scripts using strace for detailed system call tracing and performance profiling. Tools for understanding program behavior at the kernel interface level.
- **My Docker** (C, Linux, Namespaces, cgroups, seccomp) – Built a lightweight container runtime in C, using Linux namespaces, pivot_root, capabilities dropping, seccomp syscall filtering, and cgroups. Pulls and mounts a root filesystem from a container registry.
- **MyHTTPd - HTTP Server** (C, Network Programming, epoll, I/O Multiplexing) – HTTP server implementation in C with epoll-based I/O multiplexing for efficient concurrent connection handling. Features modular architecture supporting standard HTTP operations and connection management.

Education

- **Scientific Baccalaureate (Engineering & Computer Science) — Honors**, Lycée Chaptal 2016 – 2019
- **MSc Computer Engineering, Low-level & Systems (GISTRE)**, EPITA 2020 – 2026
- **Preparatory Class (PTSI) in Engineering Science**, Lycée Jules Ferry 2019 – 2020

Technical Expertise

- **Programming & Languages:** C, ASM, Shell, VHDL, CUDA, ADA, Rust, OOP.
- **Embedded & Hardware:** .
- **Other** / **DevOps:** Git, Perforce, Kubernetes, Security.